

HOST BRAIN: LINUX MPU (Cognitive Cortex)

0.5–60 Hz · Best-Effort Scheduling · Untrusted Proposals

Ingestion & Spatial Tokenizer (60 Hz)

MIPI CSI-2 DMA Rings · DINOv2 / MobileNet → 3D Affordance Tokens

Temporal World Model & Frame Graph (50 Hz)

SE(3) Lie Algebra Tree · Proprioceptive Gate · Latent Belief

Semantic Reasoning & Deliberation (1–2 Hz)

Edge VLM (PaliGemma / SmolVLM)
→ 3D Bounding Goal · Expiring Lease

Trajectory Planning & Action Chunking (20 Hz)

Diffusion / ACT → $H = 16$ Step Chunks · C^2 Spline Ensembling

Propose p_t

REFLEX BRAIN: REAL-TIME MCU (Safety Enforcer)

1000 Hz Hard Real-Time Loop · Deterministic · Zero Malloc

Control Barrier Invariants ($h(x) \geq 0$)

Zero-Allocation 1 kHz QP · Workspace
Geofence · Payload & Force Bounds

Dynamic Stopping Physics (d_{stop})

$$d_{\text{stop}}(t) = v(t) \cdot t_{\text{delay}} + \frac{v(t)^2}{2a_{\text{max}}} \leq d_{\text{clearance}} \cdot \text{Veto Accelerations}$$

Heartbeat Supervisor & Watchdog Lease

Hardware Countdown Timer · $\tau_{\text{watchdog}} \leq 50 \text{ ms}$ · Trips on Linux Crash

ISO 13850 Fallback Escalation FSM

Cat 2 (Position Hold) → Cat 1 (Dynamic Brake)
→ Cat 0 (Power Relay Open)

Filter / Verify

THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY (MPU Proposes p_t ↔ MCU Permits u_t)

HETEROGENEOUS Tightly-Coupled Memory (TCM) Shared SRAM (Lock-Free Seqlocks & DMA Rings)

Proposals (MPU → MCU): Expiring Intent Leases (t_{expire}), H -step Action Splines (p_t), Model Confidence (κ)

Telemetry (MCU → MPU): Encoders, Voltages, Veto Flags, PTP Hardware Timestamps

DMA Stream

Permitted u_t

SENSOR SUITE (Environment Ingestion)

MIPI CSI-2 Vision · 6-DoF IMU · Shunt Sense · PTP Time Base

ACTUATION STAGE & SAFETY RELAYS

3-Phase Gate Drivers · BLDC Motor Stage · Emergency Cutoff Relay

THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$)

Kinetic Momentum ($\mathbf{p} = m\mathbf{v}$) · Joule Heat ($I^2 R$) · Contact Dynamics · **Zero Software Undo**